

Exam 1

MATH 2025

Instructions

- Exam length: 75 minutes. Complete the exam in class using handwritten work only. Show supporting reasoning where appropriate.
- You may use a calculator and one sheet of handwritten notes (front and back) are permitted.
- Read each prompt carefully. If you believe a question is ambiguous, ask Dr. Friedlander.
- Unless a computation is explicitly requested, focus on conceptual explanations.
- When code or output is supplied, interpret it as directed.
- Clearly label each answer. Partial credit may be awarded for well-communicated reasoning.
- One sheet of handwritten notes (front and back) is allowed.

The exam centers on data collected from a community rail trail. For each day, **volume** records the number of users counted on the trail, and **hightemp** is the day's high temperature (°F). Additional weather variables are included in the dataset but may not be used in every question.

Summary information and figures from the data and model are provided at the end of this exam. Use these summaries and Figures 1-3 as needed; you do not need to reproduce them.

Section A. Multiple Choice

Circle the single best answer for each question. Each question is worth 4 points.

1. Which step of the data analysis cycle is most closely associated with evaluating whether the simple linear regression conditions appear reasonable for this rail trail model?
 - (A) Pose a question

- (B) Obtain data
 - (C) Understand the data
 - (D) Communicate findings
2. In Figure 1, what do the scatterplot and regression line suggest about the association between `volume` and `hightemp`?
 - (A) A strong negative linear relationship
 - (B) A moderate positive linear relationship
 - (C) No apparent relationship
 - (D) A nonlinear relationship with a clear curve
 3. Which code segment would create a scatterplot of trail volume versus high temperature using `gf_point()` from the `ggformula` package?
 - (A) `gf_point(volume ~ hightemp, data = rail_trail)`
 - (B) `gf_point(hightemp, volume, data = rail_trail)`
 - (C) `gf_point(rail_trail, volume, hightemp)`
 - (D) `gf_point(~ volume + hightemp, data = rail_trail)`
 4. Figure 3 contains a null distribution for the slope of a regression model predicting `volume` from `hightemp`. Complete this sentence. The randomization distribution was most likely generated by:
 - (A) Repeatedly shuffling `hightemp` to mimic the effect of measurement error in the predictor
 - (B) Repeatedly shuffling `volume`, to simulate the null hypothesis of no relationship
 - (C) Repeatedly resampling (bootstrapping) the data with replacement to estimate variability in the slope
 - (D) Repeatedly shuffling both variables simultaneously, to preserve the correlation structure
 5. Compared with a 95% confidence interval for the mean trail volume when `hightemp` = 80°F, a 95% prediction interval for a single day at 80°F will be:
 - (A) Narrower, because individual days are less variable than the regression line
 - (B) Wider, because it must account for both model uncertainty and day-to-day variability
 - (C) The same width, because they are based on the same fitted regression line
 - (D) Uninterpretable without a theoretical t distribution assumption

Section B. Short Answer

Show all reasoning. Each question is worth 8 points.

1. **Checking conditions:** Refer to Figure 2. Which conditions of regression are assessable from this plot? For each condition, briefly note what you observe.
 2. **Using model output:** Based on the regression table, write the fitted regression equation (using rounded coefficients) and state the predicted trail volume on a day with a high temperature of 75°F. Explain what the slope means in context.
 3. **Confidence interval and hypothesis test reasoning:** What is the difference between the null distribution, used while performing a hypothesis test, and the sampling distribution, used to create a confidence interval? How are they similar?
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Section C. True/False

For each statement, circle True or False. Each question is worth 5 points.

1. **True / False:** If we swapped the roles of `volume` and `hightemp` in the regression, the value of the correlation between the two variables would stay the same.
 2. **True / False:** A small p-value from the randomization test proves that high temperatures cause increased trail usage.
 3. **True / False:** If more observations are added that follow the same linear trend, we would expect the standard error of the slope to decrease.
 4. **True / False:** Even when the equal variance condition is satisfied, prediction intervals get wider as you move further from the mean of `hightemp`.
 5. **True / False:** When the confidence interval for the slope contains zero, the corresponding hypothesis test at the same confidence level will not reject the null hypothesis.
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Section D. Evaluate a Response

Each item shows a student's written response. For each, decide whether it is *correct*, *partially correct*, or *incorrect*, and explain your reasoning in 2–3 sentences. Each question is worth 10 points.

1. Student A states, "Because the observed slope (shown as the vertical line) falls far into the tail of the randomization distribution in Figure Figure 3, the p-value must be close to 1, so there is no evidence of a relationship." Evaluate Student A's conclusion.

2. Student B writes, “We must have made a mistake in our analysis because the intercept is negative and we can’t have negative volume.” Evaluate this interpretation using information from the dataset.
3. Student C creates a different model with the data. The fit a line showing that there is a positive correlation between the high-temperature and the number of accidents and make the following claim. “Bikers are more likely to crash on days when it is hot.” Assess this claim.

Reference Materials

Summary Tables

Table 1: Selected summaries for the rail trail dataset.

n	mean_volume	sd_volume	mean_hightemp	sd_hightemp
90	375.4	127.46	68.83	13.02

Table 2: Regression output for predicting trail volume from daily high temperature.

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-17.079	59.395	-0.288	0.774	-135.115	100.956
hightemp	5.702	0.848	6.724	0.000	4.017	7.387

Table 3: Model-level summaries for the rail trail regression.

r_squared	sigma	df_residual
0.339	104.19	88

Plots

``geom_smooth()` using formula = 'y ~ x'`

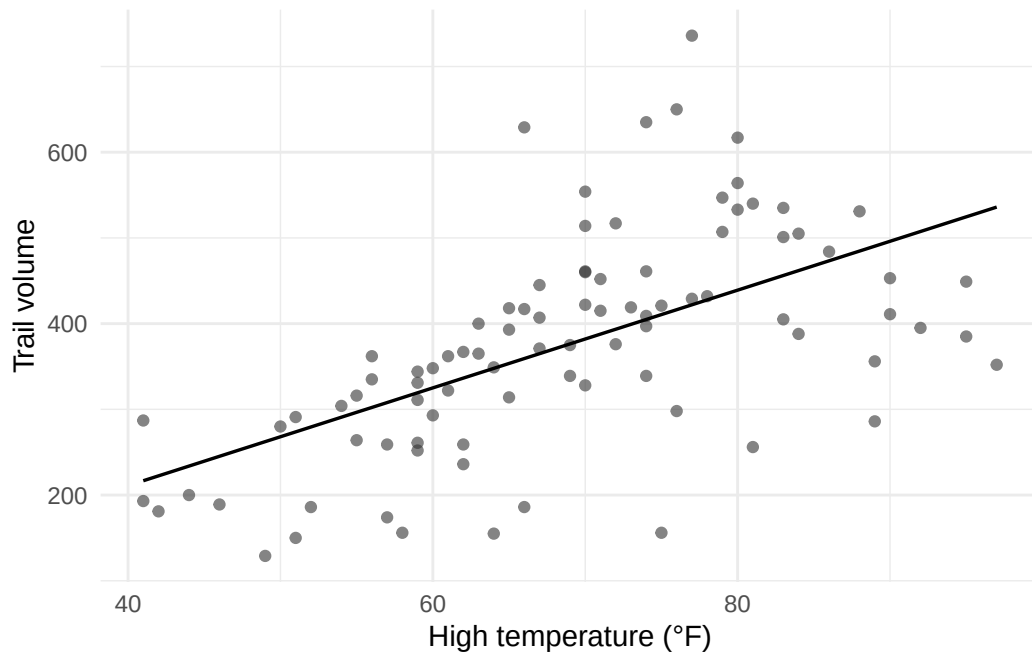


Figure 1: Trail volume vs. high temperature with fitted regression line.

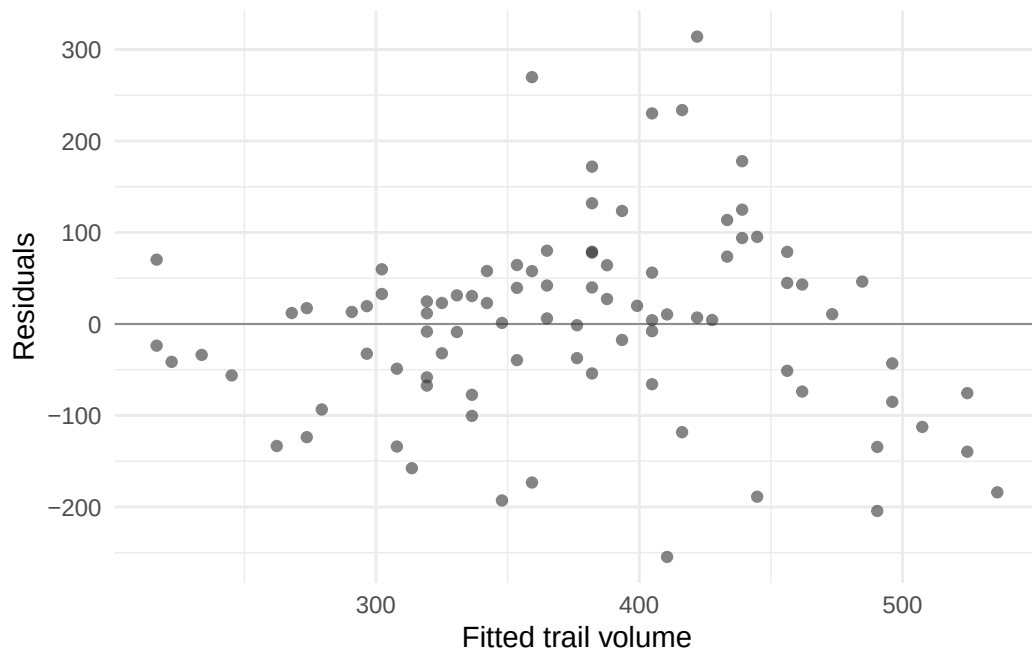


Figure 2: Residuals versus fitted values for the regression of volume on high temperature.

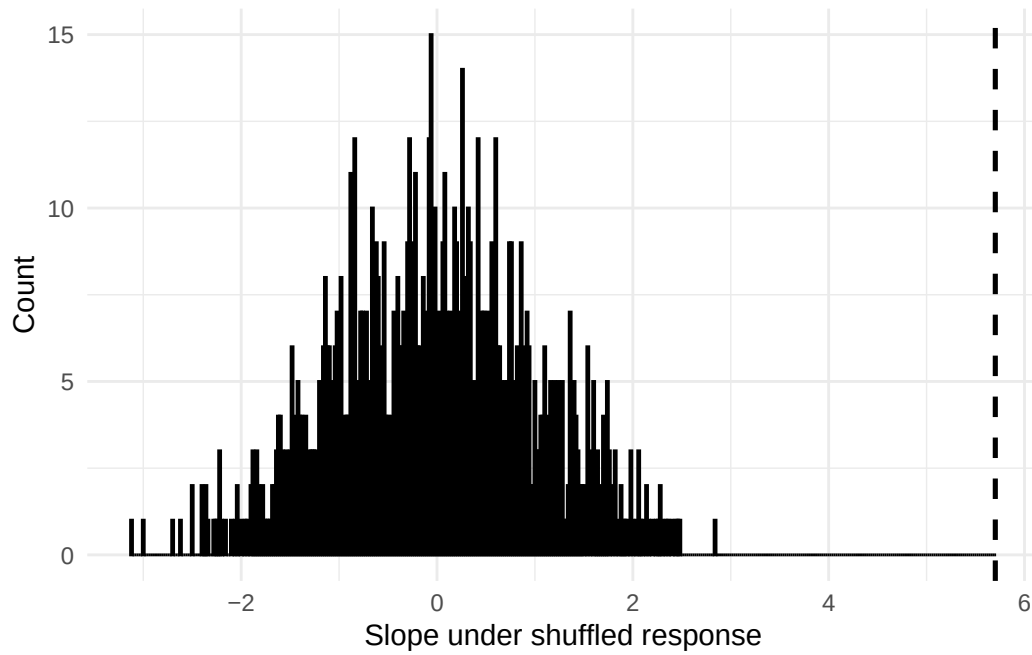


Figure 3: Randomization distribution of slope estimates assuming no relationship between temperature and trail volume.

Point Summary

Table 4: Point breakdown by section.

Section	Points
A	20
B	32
C	25
D	30
Total	107